PKM's isogeny-tower factor base in ecbench, with its auxiliary-curve search charged

2026-10-15 · `research/pkm_isogeny_tower_ecbench_20261015` · protocol committed in `43986439e` before any measured run · class: **accounting**.

Result

`prime-isogeny:t=T[,seed=S]` (`ic_framework::prime_tower`) builds PKM's isogeny tower $V = x(R' + \langle Q \rangle)$ for Q of order 2^t on an auxiliary curve $E'/\mathbb{F}_p$ found from a public seed, and charges the search (residue-class baby-step giant-step filter on every curve, Mestre's order on survivors, then Q, $\langle Q \rangle$, R', V, the Vélu chain) to the factor-base phase in group-addition equivalents. All 1,080 runs (720 measured) verified; 8 audits `OK`, 720/720 replays identical.

Measured. IC on the isogeny base costs **1.00–1.33×** IC on a plain base of equal size (paired S); the excess is the search's share of S, 0.8–25 %, set by how many curves the seed tried (3–744 at $t = 8..11$), not by curve size. On the three $v_2(p-1) = 1$ curves (P-192 shape, no Kummer tower) it is 13–25 %. Every IC arm is 17–137× rho with the negation map. Panel 3: curves tried $\approx C \cdot 2^t$ with pooled $C = 1.05$ (224 builds, $t = 8..14$), spread of 2–3 orders of magnitude between seeds.

Derived. At P-192 size ($\log_2 p = 192$, $t = 48$) the search would cost $\approx 2^{74}$ GAE (2^{82} at $t = 64$) against rho's $2^{95.8}$; the `u64` code cannot run it.



Figure 1: Paired ratios of S (95 % cluster bootstrap over 8 targets, 16 pairs) and the search's share of the isogeny arm's S. Sessions in `sessions/`, L0, counts only.

Paired ratios (ecbench compare)

curve	isogeny ÷ plain	isogeny ÷ Kummer	aux share	isogeny ÷ rho	rootfind ÷ mitm	A/A
p20 `icv1-fp20-tm728-d215a2a1`	1.035 [1.005, 1.064]	1.031 [1.006, 1.056]	5.1 %	38.5	1.196	1.000
p22 `icv1-fp22-tm2947-e15fed69`	1.140 [1.134, 1.147]	1.196 [1.188, 1.204]	12.4 %	44.4	1.115	1.000
p24 `icv1-fp24-t2705-27cbb01b`	1.002 [0.979, 1.028]	0.976 [0.951, 1.002]	1.3 %	29.6	1.078	1.000
p26 `icv1-fp26-t2838-8e779ca8`	1.078 [1.070, 1.085]	1.115 [1.097, 1.129]	7.4 %	113.5	1.044	1.000

p28 `icv1-fp28-t4155-50f492b1`	1.008 [1.004, 1.011]	1.050 [1.047, 1.054]	0.8 %	136.5	1.025	1.000
q20 `icv1-fp20-t199-d1684bf5`	1.204 [1.190, 1.218]	—	17.4 %	21.0	1.168	1.000
q22 `icv1-fp22-t285-9fc7e3e5`	1.331 [1.321, 1.343]	—	25.0 %	52.2	1.090	1.000
q24 `icv1-fp24-t4451-d5368b2d`	1.172 [1.142, 1.206]	—	13.5 %	38.3	1.058	1.000

Every S is a lower bound (prime-curve Legendre symbols, square roots, lookups and row operations counted, unpriced) for all IC arms alike. No wall time is reported (L0, shared host).



Figure 2: Mean S per arm, 16 measured runs each, every run verified.

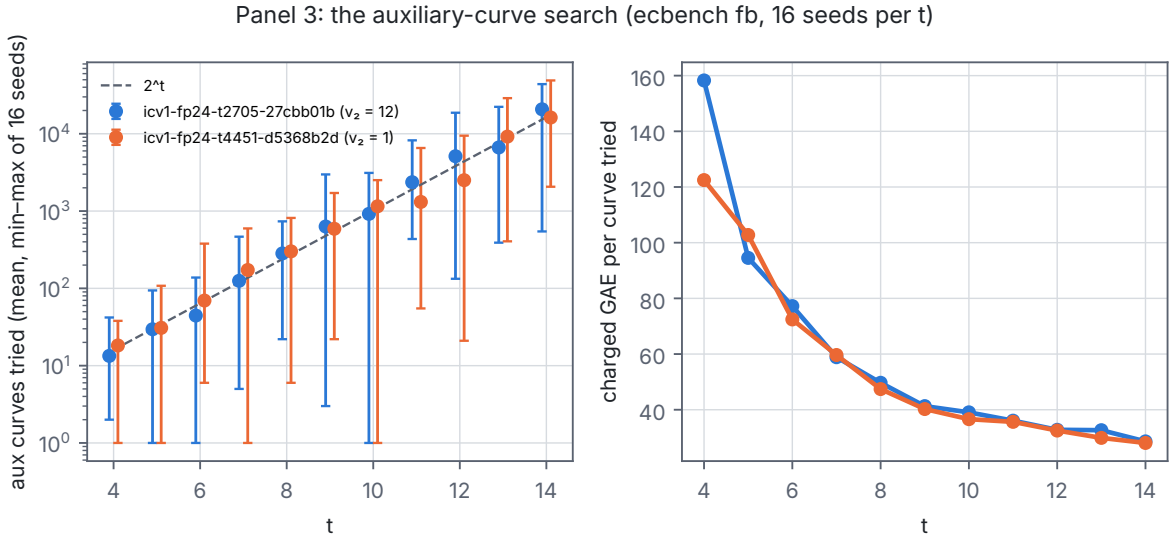


Figure 3: Panel 3: `ecbench fb` builds of `prime-isogeny:t=T,seed=S`, 16 seeds per t, two 24-bit curves.

Decision against the protocol

- P1 held (all verified, all replays identical). P2 held (tests; run `fb_id` = dump `fb_id` on all 8 curves).
- P3 failed on single draws (p20, p24, p28: tried/2^t = 0.09, 0.006, 0.05), held for the 16-seed mean at every t (0.61–1.27).
- P4 held on 7 of 8 curves; p24 not resolved (1.3 % share).
- P5 held by the letter (5.1 % ⟶ 0.8 %), not as a trend: the share follows the seed's draw.
- P6 held on 4 of 5 (p24: 0.976, Kummer base 4 abscissae larger). P7 held.
- Controls: A/A 1.000; rootfind ÷ mitm reproduces the 2026-10-13 values (1.212 … 1.026).

Projection (derived)

GAE per curve $\approx 1.5(b-t) + t + 2 \cdot 2^{(b/2+2-t)/2}$ (matches panel 3: 46 vs 50 at $t = 8$, 30 vs 29 at $t = 14$). With $C \cdot 2^t$ curves: P-192, $t = 48$: $2^{48.1}$ curves × 2^{26} = $\approx 2^{74}$ GAE, baby table 2^{25}; $t = 64$: $\approx 2^{82}$. Rho: $\sqrt{\pi r/4} \approx 2^{95.8}$.

Not done

Wide integers ($p < 2^{62}$, $t \leq 20$); SEA, CM construction of E', 2-Sylow by halving, $X_1(8)$ families; Vélu evaluations uncharged ($2t$ inversions); seed variance not in the per-curve intervals (panel 3 is the distribution); two rounds instead of three, L0 instead of L2; scoreboard and encyclopedia not updated. Conductor control plane unreachable.